3. Requirements Gathering

Project: Software-Defined Industrial Visual Anomaly Detection System: Cloud MLOps Pipeline & Simulation
Supervisor: Randa Hesiam

This section is derived from the Project Proposal. Because the production line and its users are simulated, stakeholder needs are modelled from the problem description in the proposal rather than from interviews. Items marked *(assumption)* are to be confirmed with the supervisor. Priority uses MoSCoW: M = Must, S = Should, C = Could.

3.1 Stakeholder Analysis

Stakeholder	Role / interest	Main needs	Influence	Interest
Quality inspector / QA engineer (simulated end user)	Reviews verdicts and defect locations on the dashboard	Reliable PASS/FAIL; no defective unit passes; clear boxes, class and confidence; low false rejects	Medium	High
Line operator / production supervisor (simulated)	Watches the live line and reacts to alerts	Real-time feed; fast verdict (< 1 s); simple status indicators; alerts by e-mail	Medium	High
Plant / production manager (simulated)	Cares about quality, scrap and cost	Defect rate per material and batch; trends; low running cost; audit trail	High	Medium
ML / Modeling team (Saeed Ali, Mostafa Mohamed)	Trains, evaluates and exports models	Clean, leak-free datasets; reproducible experiments; retraining and quality-gate logic	High	High
Data team (Malek Mohamed, Abdelrahman Ramadan)	Curates, labels and versions data	Dataset register; accurate box conversion; consistent preprocessing; review-queue images to label	High	High
Deployment / DevOps team (Seif Mohamed, Ahmed Khaled)	Builds and operates the AWS pipeline, simulator and dashboard	Least-privilege IAM; deployment scripts; monitoring, alarms, rollback; cost control	High	High
Supervisor: Randa Hesiam, and examiners	Evaluate the project and defense	Meets scope and KPIs; evidence (ablations, reports); a stable live demo with fallback	High	High
Dataset owners / licensors (MVTec, NEU, DAGM, Kaggle/Roboflow)	Own the public data	Respect of licences (non-commercial research); correct citation	Low	Low
Cloud provider (AWS)	Hosts the services	Usage within budget and service limits	Low	Low

Engagement summary. High-influence, high-interest stakeholders (the three teams and the supervisor) are involved through weekly syncs, milestone reviews and the final defense. End-user roles (inspector, operator, manager) are represented by the dashboard design and the usability proxy check in the KPIs. Licensors are handled through the licence register in M1.

3.2 User Stories & Use Cases

User Stories

ID	User story	Acceptance criteria	Pri.
US-01	As a QA engineer, I want every image to get a PASS/FAIL verdict so that no defective unit is shipped.	Any image with a real defect is FAILed (image-level recall >= 95% on critical classes); verdict stored for 100% of images.	M
US-02	As a QA engineer, I want to see bounding boxes, class and confidence on each image so that I can confirm the defect.	Dashboard shows annotated image via presigned URL with class, confidence and PASS/FAIL badge.	M

ID	User story	Acceptance criteria	Pri.
US-03	As a line operator, I want the verdict in under one second so that the line is not slowed down.	Warm end-to-end latency p95 < 1.0 s (S3 event to DynamoDB write).	M
US-04	As a line operator, I want an alert when defect rate or image quality changes so that I can react early.	Drift alarm triggers SNS e-mail (defect-rate, confidence, input, performance drift).	M
US-05	As a plant manager, I want defect statistics per material, class and batch so that I can spot quality trends.	Analytics panel with filters, verdicts over time and confidence distribution.	S
US-06	As an auditor, I want every decision traced to model, dataset and preprocessing versions so that results are reproducible.	DynamoDB record contains all three versions plus git-linked lineage; audit coverage 100%.	M
US-07	As a data engineer, I want uncertain images collected in a review queue so that I can label them and improve the model.	Images in the uncertain confidence band are copied to review-queue/.	S
US-08	As an ML engineer, I want retraining with an automatic quality gate so that a worse model is never deployed.	Gate checks recall, false-reject, mAP, per-material regression and latency before registration.	M
US-09	As a DevOps engineer, I want automatic rollback when a new model fails its smoke test so that production stays safe.	Failed smoke test restores the previous endpoint config and sends an SNS notification.	M
US-10	As a presenter, I want to test the system with a simulated line (stream, burst, demo, warm-up) so that I can run a stable defense demo.	Simulator modes work from YAML config; endpoint status shows Cold / Warming / Warm; fallback dataset ready.	M
US-11	As a first-time viewer, I want to read a verdict, latency and model version without help so that the dashboard is usable.	Usability check passes (target TBD by team).	C

Use Cases

Use-case actors: **Simulator** (replaces the camera), **Inspector/Operator**, **Data/ML engineer**, **DevOps engineer**, **Presenter**.

UC-01: Inspect an image automatically	
Actor	Simulator (primary); AWS pipeline
Precondition	Endpoint deployed; model, thresholds and preprocessing versions configured.
Main flow	1. Simulator uploads image to raw-images/{material}/{batch}/{id}.png with metadata. 2. S3 event triggers the inference Lambda. 3. Lambda validates the file and applies shared preprocess.py. 4. Lambda calls the SageMaker Serverless endpoint (YOLOv8 ONNX). 5. Lambda applies the material threshold to get PASS/FAIL. 6. Lambda writes the record to DynamoDB and publishes drift metrics.
Alternative flows	A1 Invalid file: rejected and logged, DLQ. A2 Endpoint error or timeout: retry, then DLQ. A3 Cold start: flagged cold_start=true and reported separately. A4 Warm-up image: no record written.
Postcondition	One audit record exists with verdict, detections, latency and versions.

UC-02: Monitor the live line on the dashboard	
Actor	Inspector / Operator
Precondition	Pipeline running; records present in DynamoDB.
Main flow	1. User opens the dashboard. 2. Live feed shows annotated images and PASS/FAIL badges. 3. User filters by material, class or verdict. 4. User checks latency (p50/p95/p99), error rate and throughput. 5. User checks active model and live recall / false-reject vs ground truth.
Alternative flows	A1 Endpoint cold: status shows Cold/Warming and user presses Warm-up. A2 No data: dashboard shows an empty-state message.
Postcondition	User knows the line status and current model version.

UC-03: Handle a drift alarm and the review queue	
Actor	Operator; Data engineer
Precondition	Drift metrics and alarms deployed (M4).
Main flow	1. CloudWatch alarm fires (defect-rate, confidence, input or performance drift). 2. SNS e-mail is sent. 3. Uncertain images are in review-queue/. 4. Data engineer reviews and labels them in Roboflow. 5. A new dataset version is published.
Alternative flows	A1 False alarm: engineer closes the case and records it. A2 New defect class found: marked as held-out/new class for retraining.
Postcondition	A new labelled dataset version is ready for retraining.

UC-04: Retrain, gate and deploy a new model	
Actor	ML engineer; Supervisor/approver; DevOps engineer
Precondition	New dataset version published; trigger event sent to EventBridge.
Main flow	1. Pull dataset version to S3. 2. Train YOLOv8 with pinned preprocess.py and mask_to_boxes.py. 3. Evaluate on previous and updated test sets; export ONNX; parity check. 4. Quality gate. 5. Register as PendingManualApproval and notify by SNS. 6. Approver approves. 7. Deployment Lambda updates the endpoint (blue/green), warms up, runs smoke test on golden set.
Alternative flows	A1 Gate fails: model not registered. A2 Approval rejected: stays in registry, not deployed. A3 Smoke test fails: automatic rollback and SNS notification.
Postcondition	New model is live, or the previous one is restored; version recorded.

UC-05: Run a simulated line and defense demo	
Actor	Presenter; Simulator
Precondition	Keep-warm enabled for test/defense; fallback dataset prepared.
Main flow	1. Presenter runs simulator --warmup N. 2. Dashboard shows status Warm and stable p95. 3. Presenter runs steady, burst or scripted demo mode with a chosen defect mix. 4. Dashboard shows results; held-out-class scenario shows drift and before/after.
Alternative flows	A1 Live system fails: switch to fixed demo dataset and pre-recorded results.
Postcondition	Demo completed; keep-warm disabled afterwards for cost control.

3.3 Functional Requirements

ID	Requirement	Pri.	Source / Milestone
FR-01	The simulator shall replay images from the held-out test split to S3 with the key pattern raw-images/{material}/{batch_id}/{image_id}.png and metadata (material, batch, simulated timestamp).	M	US-10 / M4
FR-02	The simulator shall support steady stream (1-5 images/s), burst, scripted demo (chosen defect mix) and --warmup N modes, configured through YAML.	M	US-10 / M4
FR-03	The simulator shall produce a ground-truth manifest so live recall and false-reject rate can be measured.	M	US-06 / M4

ID	Requirement	Pri.	Source / Milestone
FR-04	An S3 ObjectCreated event shall trigger the inference Lambda automatically, with no manual step.	M	US-01 / M4
FR-05	The Lambda shall validate each file (format, size, readable) and reject invalid files to a DLQ.	M	US-01 / M4
FR-06	The system shall apply the shared, versioned preprocess.py (letterbox 640x640, grayscale, material profile) identically in training and serving.	M	US-06 / M1
FR-07	The system shall detect and localise defects of the classes crack, scratch, chip, pit and stain/rust with bounding boxes, class label and confidence (0-1).	M	US-02 / M2
FR-08	The system shall return PASS if no detection exceeds the material-specific threshold, otherwise FAIL, and record the maximum score.	M	US-01 / M2
FR-09	The system shall support two material families (ceramics, metals) with separate preprocessing profiles and thresholds.	M	US-01 / M1
FR-10	The system shall store one DynamoDB record per inspection (image_id, timestamp, material, batch, verdict, max_confidence, detections, latency_ms, cold_start, model/dataset/preprocess versions, input_stats, s3_uri, optional ground_truth).	M	US-06 / M4
FR-11	Writes shall be idempotent, keyed by image_id.	M	US-06 / M4
FR-12	Warm-up requests (warmup=true) shall not create inspection records and shall be excluded from KPIs and the dashboard.	M	US-10 / M3
FR-13	The system shall publish custom CloudWatch metrics per inspection and raise alarms for defect-rate, confidence, input and performance drift, with SNS e-mail notification.	M	US-04 / M4
FR-14	Images in the uncertain confidence band shall be copied to the review-queue/ prefix.	S	US-07 / M4
FR-15	The dashboard shall show a live feed of annotated images (presigned URLs) with class, confidence and PASS/FAIL badge.	M	US-02 / M5
FR-16	The dashboard shall provide analytics: defect rate per material and class, verdicts over time, confidence distribution and filters.	S	US-05 / M5
FR-17	The dashboard shall provide a performance view: latency p50/p95/p99, error rate, throughput, separate cold-start panel and a 1-second marker.	M	US-03 / M5
FR-18	The dashboard shall provide a model panel (active model, dataset and preprocessing versions, thresholds, live recall / false-reject) and an MLOps panel (drift indicators, review-queue size, last retraining, gate result, before/after).	M	US-06, US-08 / M5
FR-19	The dashboard shall show endpoint status (Cold / Warming / Warm) and a Warm-up button.	M	US-10 / M5
FR-20	An EventBridge schedule shall keep the endpoint and inference Lambda warm during test and defense phases only.	S	US-10 / M3-M5
FR-21	A scripted retraining workflow shall pull a dataset version, train, evaluate on previous and updated test sets, export ONNX and run a parity check.	M	US-08 / M5
FR-22	A quality gate shall require: critical image-level recall >= 95%, false-reject <= 5%, mAP not worse beyond margin, no material regression, recall on any new class meets target, and latency within budget.	M	US-08 / M5
FR-23	Passing models shall be registered in SageMaker Model Registry as PendingManualApproval with dataset version, git commit and preprocessing version, and an SNS notification sent.	M	US-08 / M5
FR-24	On approval, a deployment Lambda shall update the endpoint (blue/green), warm it up, run a smoke test on a golden set and confirm.	M	US-09 / M5
FR-25	If the smoke test fails, the system shall automatically roll back to the previous endpoint config and send an SNS notification.	M	US-09 / M5
FR-26	The data pipeline shall convert masks to boxes with mask_to_boxes.py (binarise, merge, components, noise filter, padding, minimum size, diagonal and DAGM-ellipse rules) and pass the five unit-test cases.	M	US-01 / M1

ID	Requirement	Pri.	Source / Milestone
FR-27	The dataset shall be split 70/20/10 by source-image group, include negatives from every source, and have duplicates removed.	M	US-01 / M1
FR-28	The system shall evaluate and report results per source, per material and by defect size (small/medium/large).	M	US-01 / M2

3.4 Non-functional Requirements

ID	Category	Requirement	Measure / target
NFR-01	Performance	Warm end-to-end latency per image	p50/p95/p99 reported; < 1.0 s (S3 event to DynamoDB write)
NFR-02	Performance	Cold-start latency measured and reported separately; idle time to cold measured at 5/10/15/30 min	Reported KPI; near-warm on defense day
NFR-03	Performance	Throughput sustains simulator rate; bursts tested	1-5 images/s steady; burst test in M3/M4
NFR-04	Accuracy	Detection quality on held-out test set	Image-level recall >= 95%; mAP@0.5 >= 0.90; false-reject <= 5%
NFR-05	Reliability	Pipeline failures captured, not lost	DLQ, S3 event retries, idempotent writes; low pipeline error rate
NFR-06	Reliability	Stable during test and defense	No failed demo run (target TBD by team); recorded fallback available
NFR-07	Reliability	Safe deployment	Automatic rollback on failed smoke test; no-downtime blue/green update
NFR-08	Security	Access control and secrets	Least-privilege IAM per service; no hard-coded credentials; CloudTrail enabled
NFR-09	Security	Data protection	Private buckets; encryption at rest (SSE-KMS) and in transit
NFR-10	Usability	Dashboard is self-explanatory	Full demo without developer help; first-time viewer can read verdict, latency and model version (check TBD)
NFR-11	Usability	Clear visual feedback	Colour-coded PASS/FAIL badge; Cold/Warming/Warm indicator
NFR-12	Traceability	Every decision auditable	Audit coverage 100%: model, dataset and preprocessing versions in each record
NFR-13	Reproducibility	Training and serving consistency	Single versioned preprocess.py; ONNX parity check; every model linked to dataset version, preprocessing version and git commit
NFR-14	Maintainability	Code quality and tests	Unit tests for preprocess.py and mask_to_boxes.py; documented runbooks; Git hygiene
NFR-15	Scalability	Handle load changes without manual scaling	Serverless pay-per-request services; endpoint concurrency sized to simulator rate
NFR-16	Cost	Stay within budget	AWS Budgets alarms, small model, free Colab/Kaggle GPU or Spot, keep-warm only for test/defense, teardown scripts; total AWS budget 60 USD
NFR-17	Portability	Workflow not tied to one orchestrator	Gate and registry metadata compatible with SageMaker Pipelines (future work)
NFR-18	Compliance	Respect dataset licences	Licence register in M1; non-commercial academic use; sources cited
NFR-19	Extensibility	Add new materials without redesign	Material-profile design allows a glass profile later

3.5 Assumptions and Constraints

- The line, camera and users are simulated; no physical hardware or factory access.
- Data is public (MVTec AD, NEU, DAGM, one Kaggle/Roboflow ceramics set); results demonstrate the method, not guaranteed performance on a real line.
- Six-week schedule (18 Oct - 28 Nov 2026); scope limited to ceramics and metals; glass, OBB boxes and SageMaker Pipelines are future work.
- YOLOv8 cannot detect defect types absent from training data; such cases are handled by the review queue and retraining loop.
- Model approval and labelling are manual (human-in-the-loop); training, gate and deployment are automated.
- Stakeholder roles for end users (inspector, operator, manager) are modelled *(assumption)* and to be confirmed with the supervisor.